

Practical2 : Configuring the CI/CD Pipeline for Telecom Tower Failure Prediction using AWS CodePipeline

Step 1 – Create a GitHub Repository

Objective: Store the Machine Learning project in a central repository so AWS can access the latest code.

Procedure

1. Open **GitHub** and log in.
2. Click **New Repository**.
3. Repository Name:

Telecom-MLOps

4. Upload the following files:

telecom_tower_failure.csv
train.py
requirements.txt
buildspec.yml
README.md

5. Click **Commit Changes**.

Step 2 – Create an Amazon S3 Bucket

Objective: Store the trained Machine Learning model (`model.pkl`) in cloud storage.

Procedure

1. Login to AWS Console.
2. Search for **Amazon S3**.
3. Click **Create Bucket**.

4. Bucket Name:

telecom-mlops-model-storage

5. Leave all settings as default.
6. Click **Create Bucket**.

Step 3 – Create an IAM Role

Objective: Give AWS CodeBuild permission to access Amazon S3.

Procedure

1. Open **IAM**.
2. Click **Roles**.
3. Click **Create Role**.
4. Select:
 - **Trusted Entity:** AWS Service
 - **Use Case:** CodeBuild
5. Attach the following policies:
 - AmazonS3FullAccess
 - AWSCodeBuildDeveloperAccess
 - CloudWatchLogsFullAccess
6. Role Name:

CodeBuildTelecomRole

7. Click **Create Role**.

Step 4 – Create the

buildspec.yml

File

Objective: Tell AWS exactly how to build and train the model.

Procedure

Create a file named:

buildspec.yml

Paste the following code:

```
version: 0.2
```

```
phases:
```

```
  install:
```

```
    runtime-versions:
```

```
      python: 3.11
```

```
  commands:
```

```
    - pip install -r requirements.txt
```

```
  build:
```

```
    commands:
```

```
      - python train.py
```

```
artifacts:
```

```
  files:
```

```
    - model.pkl
```

Save the file and push it to GitHub.

Explain to Students

“**buildspec.yml** is like an instruction manual for AWS. It tells AWS which libraries to install, which Python file to execute, and which output file (**model.pkl**) should be saved.”

Step 5 – Create an AWS CodeBuild Project

Objective: Configure AWS to automatically train the Random Forest model.

Procedure

1. Open **AWS CodeBuild**.
2. Click **Create Build Project**.

Project Name:

TelecomModelTraining

3. Source Provider:

GitHub

4. Connect your **Telecom-MLOps** repository.
5. Environment:
 - Operating System: Ubuntu
 - Runtime: Managed Image
 - Language: Python
6. Service Role:

CodeBuildTelecomRole

7. Build Specification:

Use buildspec.yml

8. Click **Create Build Project**.

Step 6 – Test the Build Process

Objective: Verify that AWS can successfully train the model.

Procedure

1. Open the **CodeBuild Project**.
2. Click **Start Build**.

AWS performs the following steps automatically:

Download Source Code
↓
Install Python Libraries
↓
Read telecom_tower_failure.csv
↓
Train Random Forest Model
↓
Generate model.pkl
↓
Build Successful

3. Open **Build History** → **Latest Build** → **View Logs**.

Step 7 – Create an AWS CodePipeline

Objective: Automate the complete Machine Learning workflow.

Procedure

1. Open **AWS CodePipeline**.
2. Click **Create Pipeline**.

Pipeline Name:

TelecomMLOpsPipeline

3. Source Stage:
 - Provider: GitHub Version 2
 - Repository: Telecom-MLOps
 - Branch: main
4. Build Stage:
 - Provider: AWS CodeBuild
 - Project: TelecomModelTraining
5. Deployment Stage:
 - Skip Deployment
6. Click **Create Pipeline**.

Step 8 – Trigger Automatic Training

Objective: Verify that the pipeline automatically retrains the model.

Procedure

Open `train.py`.

Change:

```
n_estimators = 100
```

to

```
n_estimators = 150
```

Save the file.

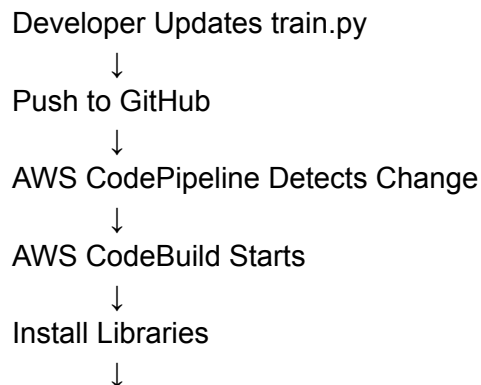
Commit the changes.

Push the updated code to GitHub.

Step 9 – Observe Automatic Pipeline Execution

After pushing the code, AWS automatically starts the pipeline.

The execution flow is:



Execute train.py



Train Random Forest Model



Generate model.pkl



Upload model.pkl to Amazon S3

Step 10 – Verify the Output

1. Open **Amazon S3**.
2. Open the bucket:

telecom-mlops-model-storage

3. Verify that the latest `model.pkl` file is present.

Expected Output

- The updated code is pushed to GitHub.
- AWS CodePipeline automatically detects the change.
- AWS CodeBuild retrains the Random Forest model.
- A new `model.pkl` file is generated.
- The trained model is uploaded to Amazon S3.
- The entire process is completed without any manual retraining.